

Tree Species Identification with Deep Learning

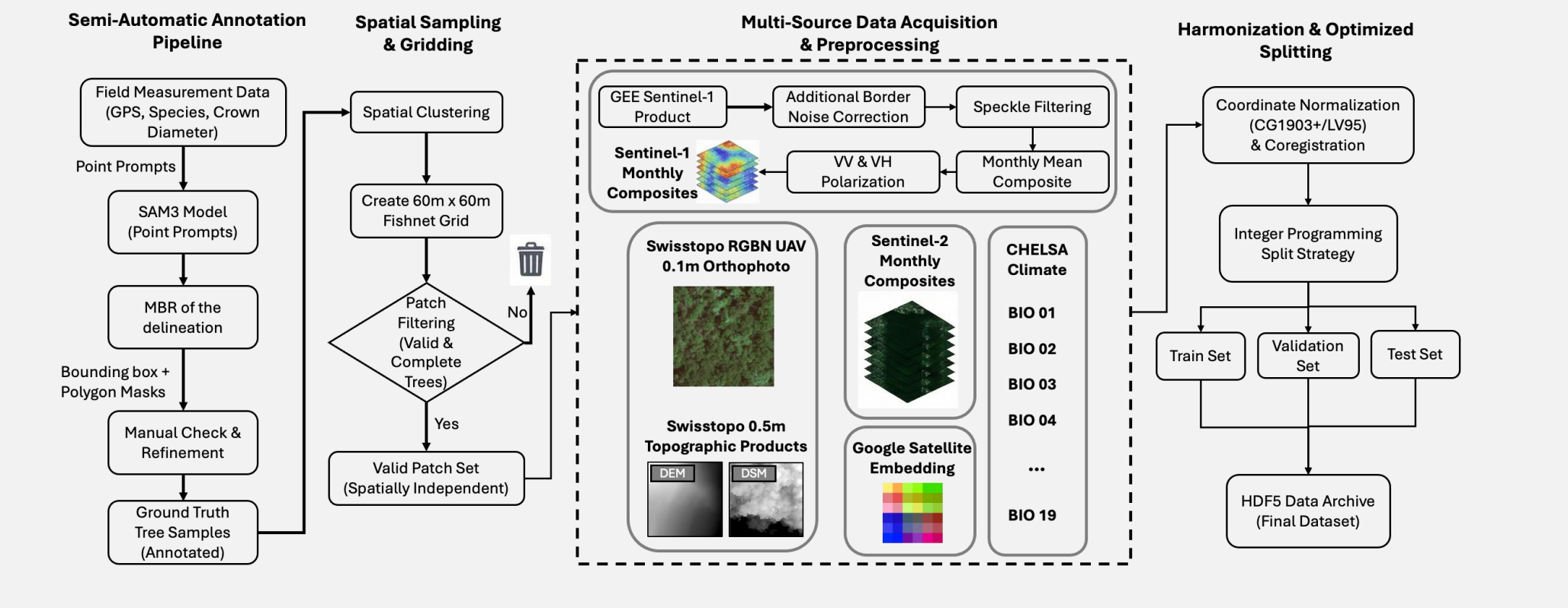
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1 Introduction

Although remote sensing enables large-scale forest analysis, tree species identification remains challenging with single-source imagery¹. This project explores how **multi-modal data integration** can enhance model performance and improve fine-grained tree species classification.

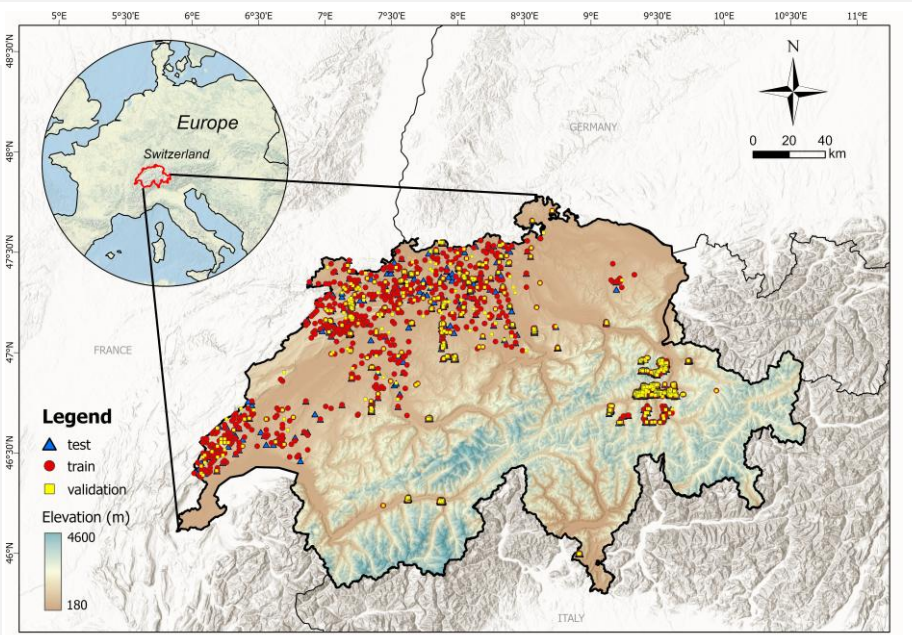
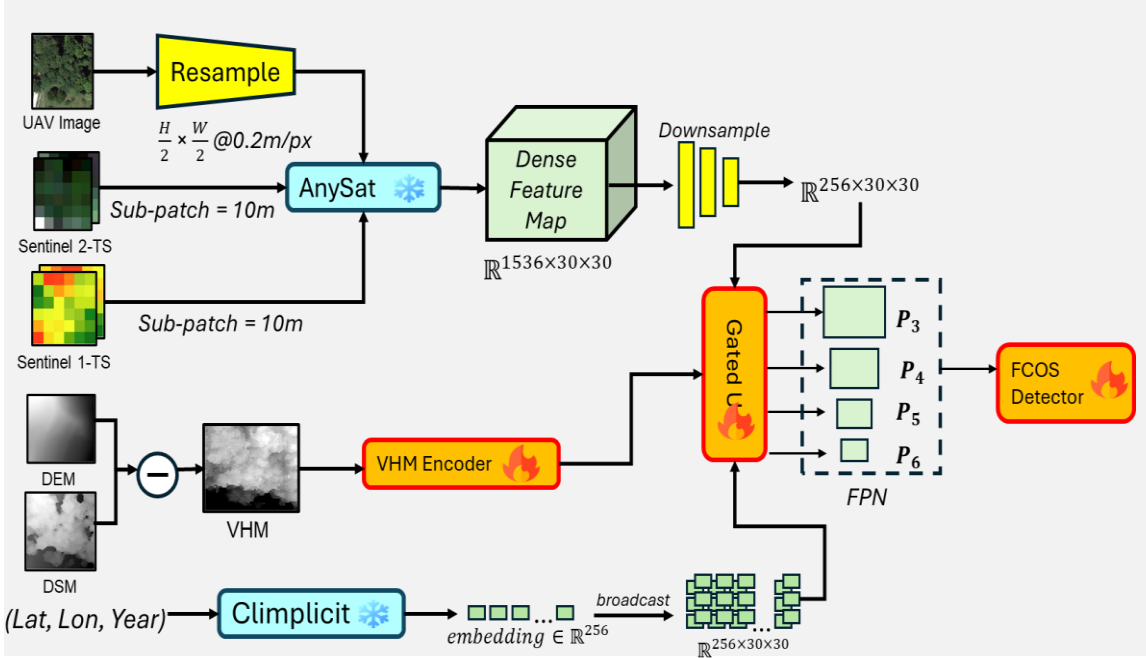
2 TreeAI-Swiss Dataset

TreeAI-Swiss is a nationwide multi-modal (RGBN image, Sentinel 1/2 time series, topographic data, CHELSA climate variables, Google satellite Embedding v1) dataset, based on on-site validated observations. It has 10 species, 5 families,



3 Model

The model follows a **multi-modal fusion design**, integrating satellite imagery, structural cues, and environmental embeddings into a unified representation through a lightweight fusion backbone.



5 Conclusion

Incorporating S1/2, VHM, and climate information steadily improves both detection accuracy and species classification, demonstrating the complementary value of multi-modal inputs.

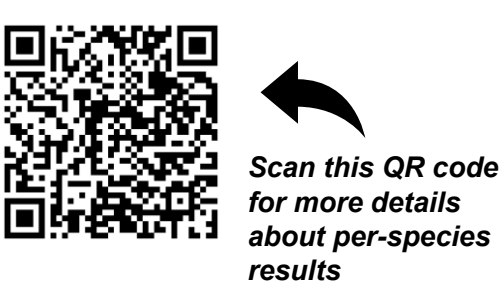
4 Results

Method Setup	Input Modality				Temporal Scale	Detection HR@0.4	Species-Genus (10 cls)		Family (5 cls)	
	RGB	S1/2	VHM	Clim.			Acc.	F1	Acc.	F1
RGB Baseline	✓				Month	0.652	0.566	0.565	0.738	0.758
+ S1/2	✓	✓			Month	0.672	0.572	0.607	0.674	0.717
+ S1/2 + VHM	✓	✓	✓		Month	0.636	0.593	0.609	0.751	0.764
+ S1/2 + Climate	✓	✓		✓	Month	0.704	0.639	0.656	0.743	0.767
Impact of Temporal Duration (with all modalities)										
Full Model (Annual)	✓	✓	✓	✓	Year	0.653	0.562	0.564	0.631	0.659
Full Model (Quarterly)	✓	✓	✓	✓	Quarter	0.659	0.559	0.581	0.722	0.744
Full Model (Monthly)	✓	✓	✓	✓	Month	0.660	0.666	0.677	0.773	0.793

Table 1. Comparison of modality configurations and temporal scales. The first section evaluates how adding S1/2, VHM, and climate modalities improves detection (HR@0.4) and species classification. The second section examines the effect of different temporal durations (annual, quarterly, monthly).

References

1. Fassnacht, F.E., Latifi, H., Stereńczak, K., Modzelewska, A., Lefsky, M., Waser, L.T., Straub, C., Ghosh, A., 2016. Review of studies on tree species classification from remotely sensed data. Remote Sensing of Environment 186, 64–87. <https://doi.org/10.1016/j.rse.2016.08.013>



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